Research Project

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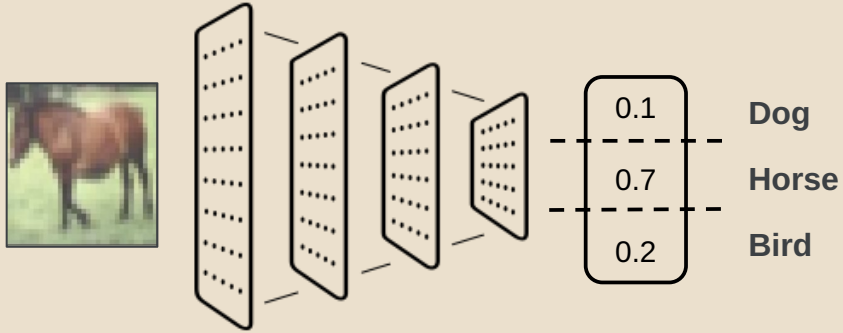
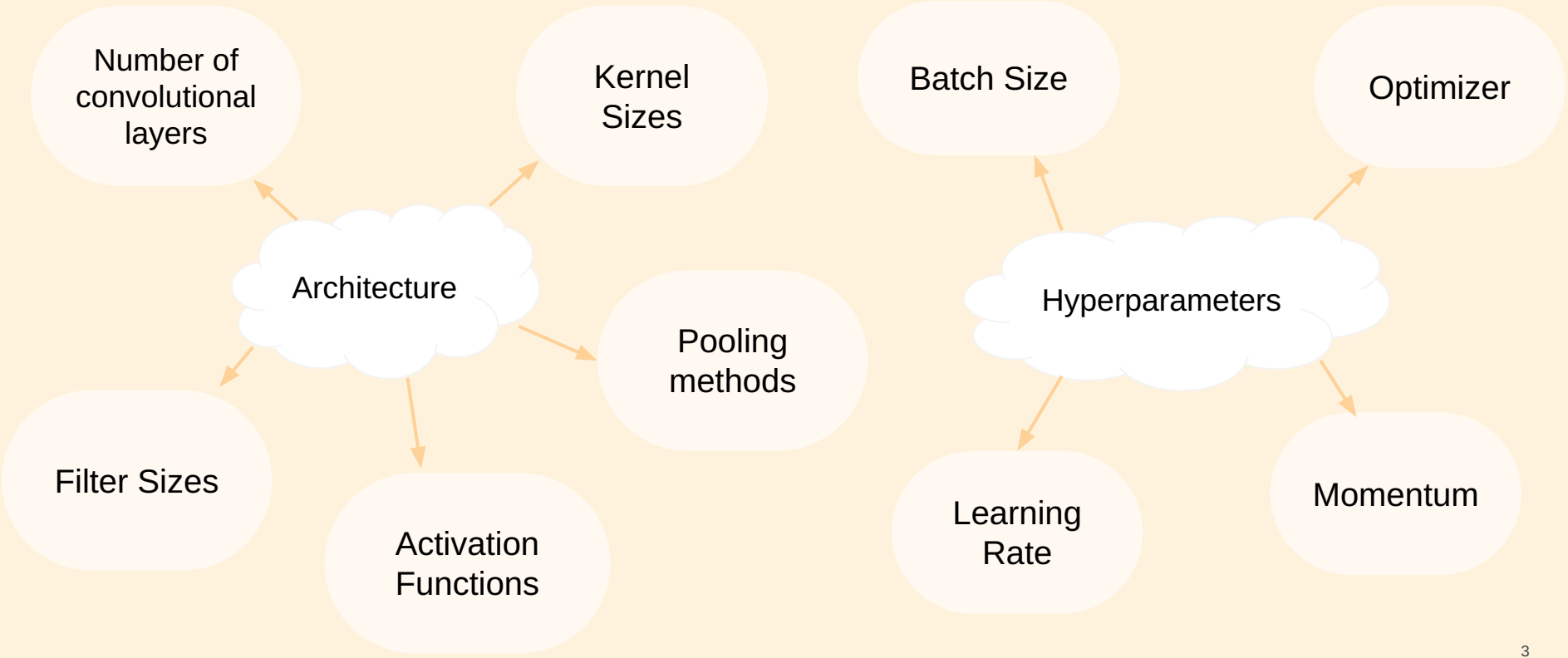
Research Goal

- Build a convolutional neural network (CNN) that can accurately classify images in the CIFAR-10 dataset



- Using CNN to preserve spatial structure for processing grid-structured data such as images in order to capture features like edges, textures, and object parts
  - 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck
  - 32 pixel images, 3 color channels (Red, Green, Blue)
- Improve model performance through architecture and hyperparameter tuning while accounting for time constraints and computational costs

Parameter Optimization



Planned Approaches

1 NAS

Systematically explores a defined space of architectural parameters, such as layer depth, kernel size, and pooling strategy, to determine the CNN configuration that maximizes validation accuracy.

2 DARTS

A gradient-based NAS approach that makes architecture search efficient by learning weighted combinations of operations and choosing subsequent architecture features based on weights.

3 Bayesian Optimization

A probabilistic hyperparameter tuning method that uses a Gaussian process to model performance and select the most promising configurations efficiently.

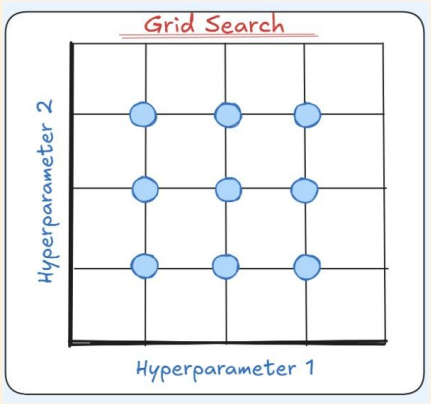
Grid/Random Search:

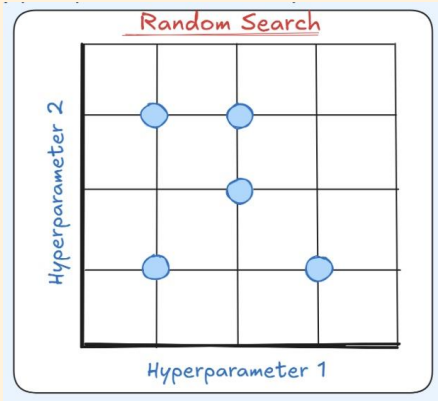
Grid search exhaustively tests all hyperparameter combinations, while random search samples a subset - often finding good configurations with lower cost.

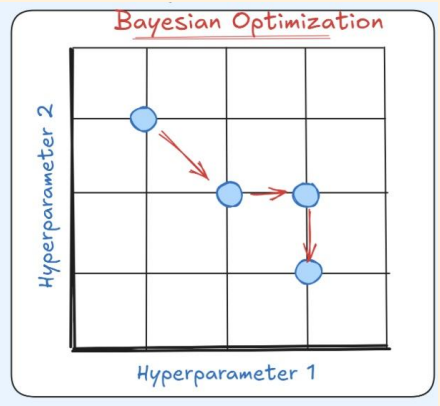
Performance Optimization Plan:

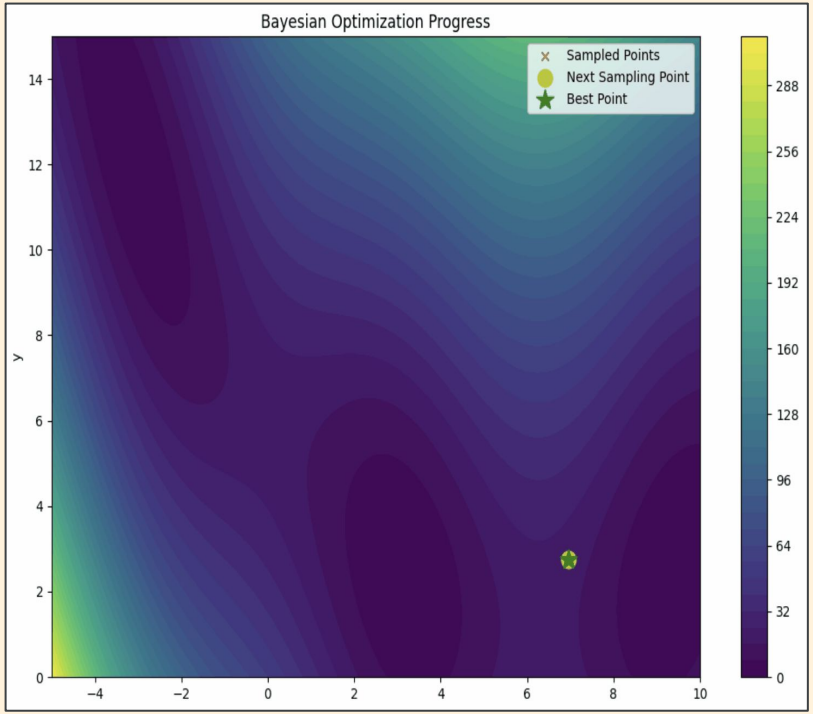
Two-stage optimization combining constrained DARTS-based architecture search and Bayesian hyperparameter tuning.

Hyperparameter Design



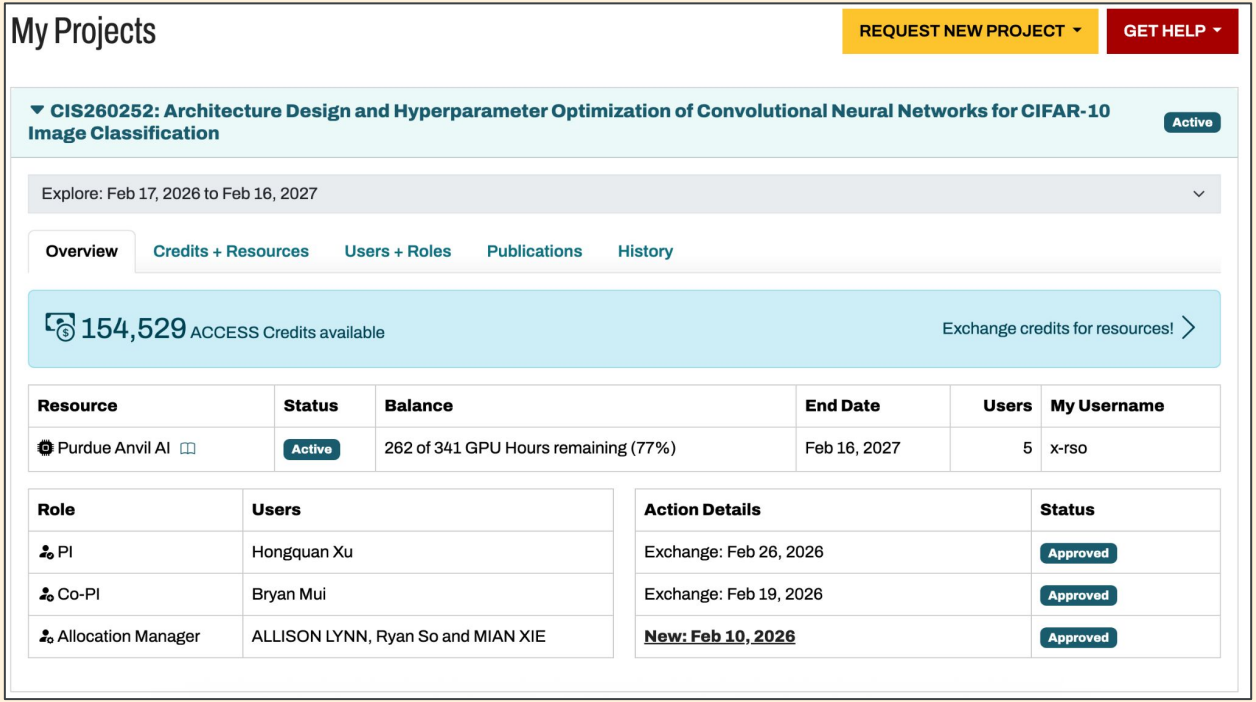






Access

- Program funded by the National Science Foundation (NSF) to provide high-performance computing (HPC) and GPU-based systems to researchers and students



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CIS260252: Architecture Design and Hyperparameter Optimization of Convolutional Neural Networks for CIFAR-10 Image Classification Active

Explore: Feb 17, 2026 to Feb 16, 2027

Overview | Credits + Resources | Users + Roles | Publications | History

154,529 ACCESS Credits available — Exchange credits for resources!

Resource	Status	Balance	End Date	Users	My Username
Purdue Anvil AI	Active	262 of 341 GPU Hours remaining (77%)	Feb 16, 2027	5	x-rso

Role	Users
PI	Hongquan Xu
Co-PI	Bryan Mui
Allocation Manager	ALLISON LYNN, Ryan So and MIAN XIE

Action Details	Status
Exchange: Feb 26, 2026	Approved
Exchange: Feb 19, 2026	Approved
New: Feb 10, 2026	Approved

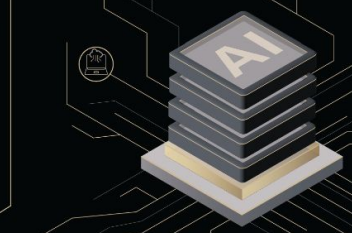
ANVIL

AnvilGPT

- ❑ Secure LLM Service
- ❑ UI & API access to open source LLMs
- ❑ Vector store for RAG pipelines

Compute & Storage

- ❑ 64x 40GB A100s
- ❑ 84x 80GB H100 SXM
- ❑ 1 PB Object Storage
- ❑ Swappable between batch and cloud



Support

- ❑ Dedicated allocation liaison
- ❑ Consultations with AI Research Scientist
- ❑ Trainings on Generative AI

Composable Cloud

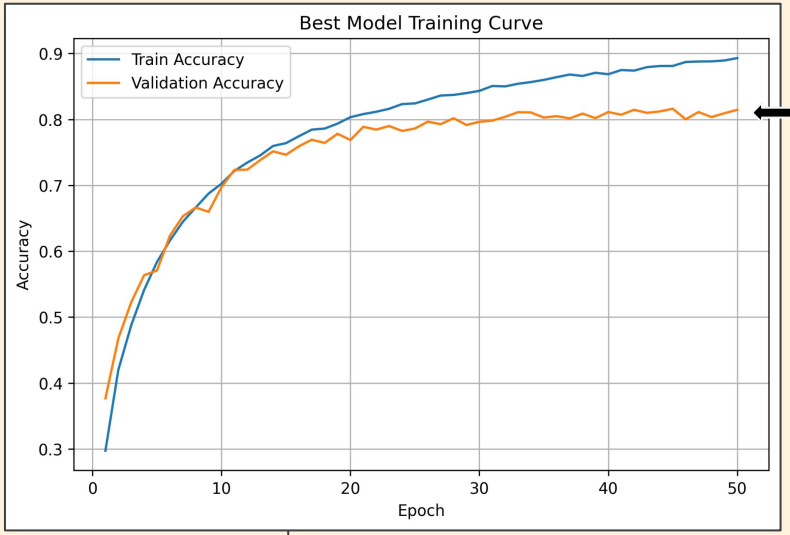
- ❑ Rancher-managed Kubernetes cluster
- ❑ MIG-enabled GPUs
- ❑ ML Inference Service Deployment
- ❑ Jupyter Notebook Service

PURDUE UNIVERSITY | Rosen Center for Advanced Computing

NSF

#2005632

Results - Architecture Selection



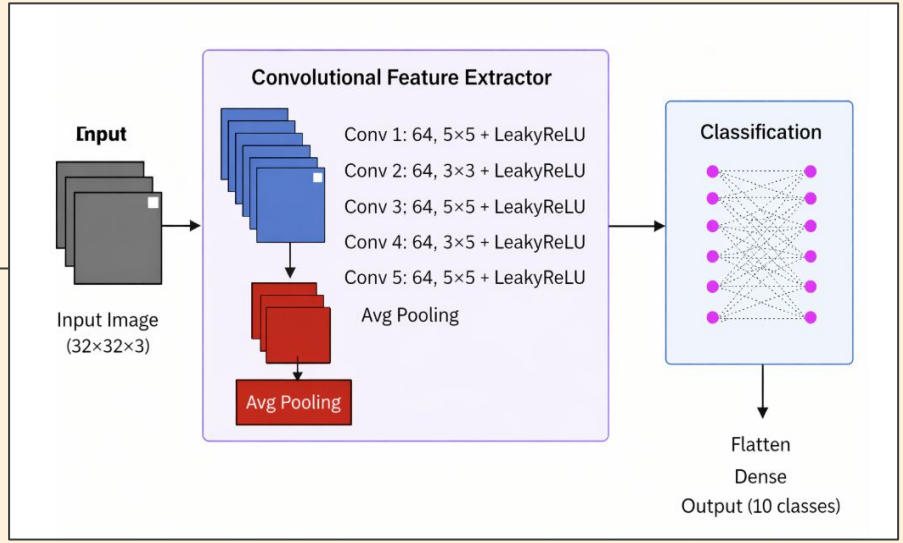
Final Val Accuracy: 81.5%

Average Train Time per Epoch: 5.12 mins

Selected Architecture

# of Layers	Kernel Sizes	Filters	Activation Function	Pooling Type
5	[5, 3, 5, 3, 5]	All 64	LeakyReLu	Avg

Best Model (Highest Validation Accuracy)

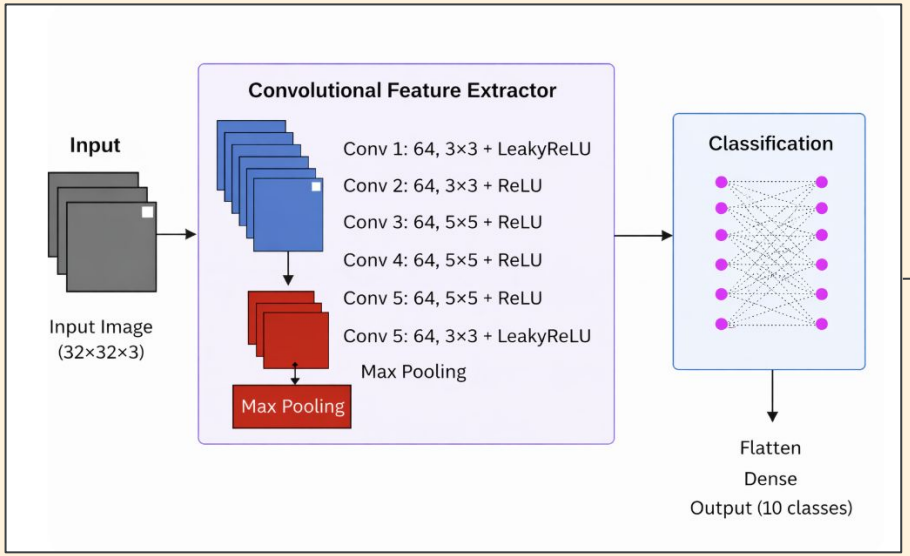


Neural Architecture Search (NAS)

50 epochs, 50 trials

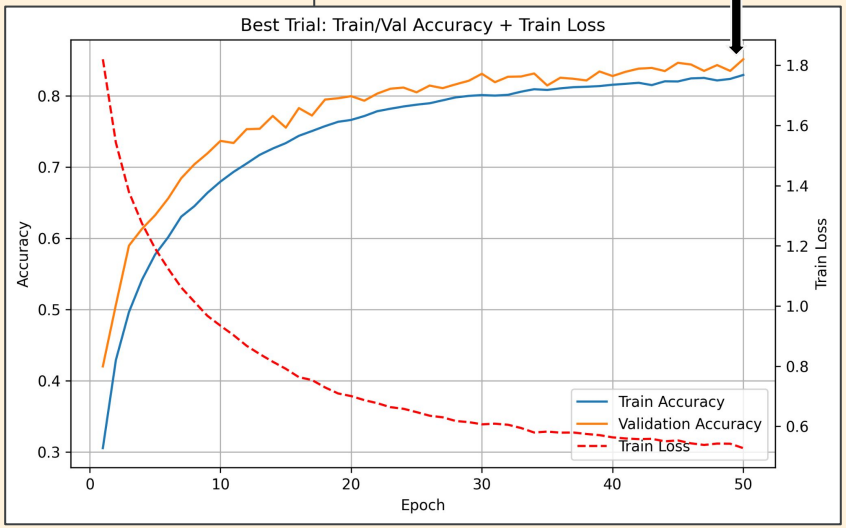
Differential Neural Architecture Search (DARTS)

50 epochs, 50 trials



Best Model (Highest Validation Accuracy)

Final Val Accuracy: 85.2% + 3.7% from NAS

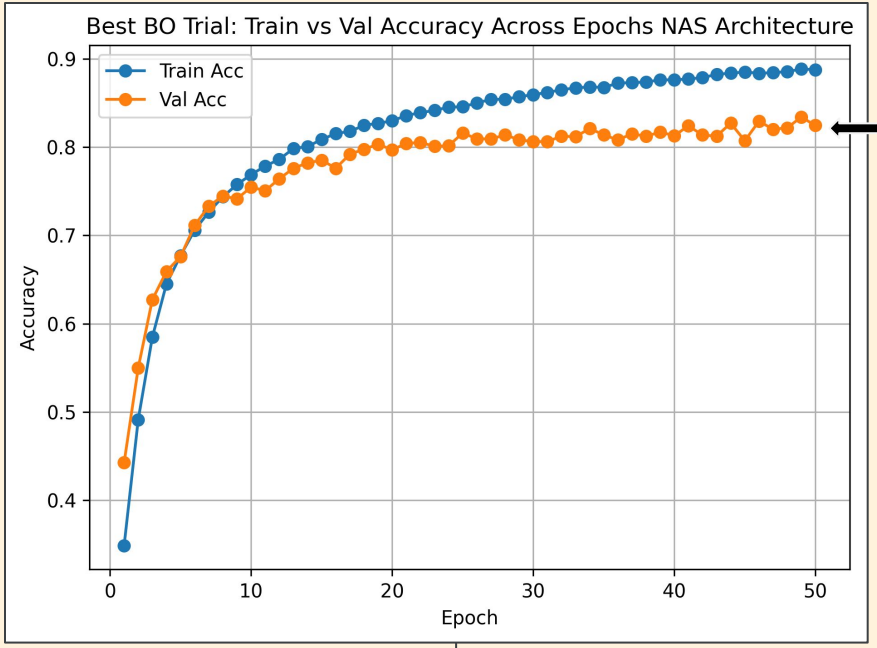


Average Train Time per Epoch: 13.61 mins

Selected Architecture

# of Layers	Kernel Sizes	Filters	Activation Function	Pooling Type
5	[3, 3, 5, 5, 3]	All 64	[LeakyReLu, ReLu, ReLu, ReLu, LeakyReLu]	Max

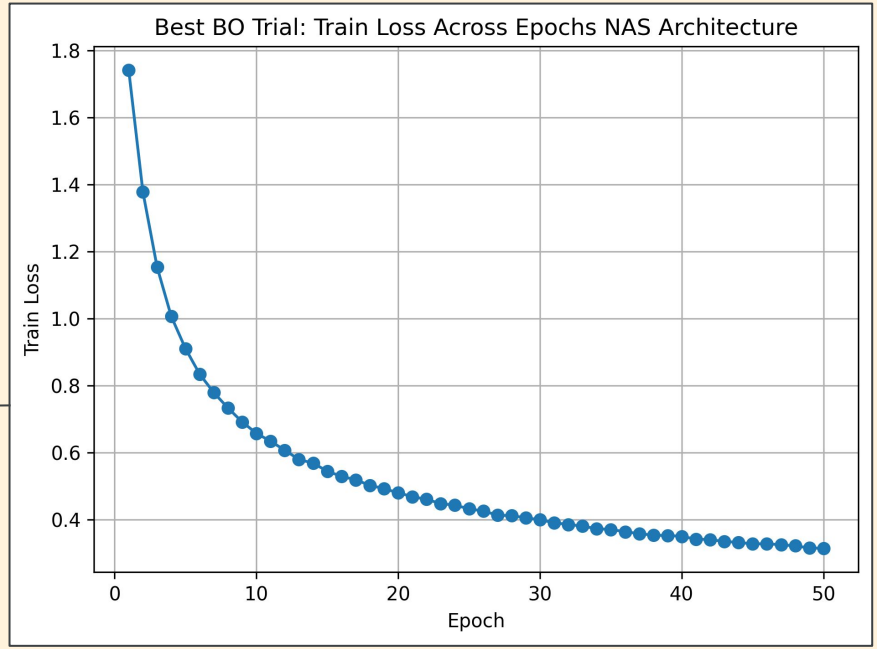
Results - Hyperparameter Tuning



Final Val Accuracy: 83.4%

Selected Hyperparameters

Learning Rate	Weight Decay	Batch Size	Optimizer	Momentum
1.238× 10−3	2.2× 10−5	32	Adam	0.857

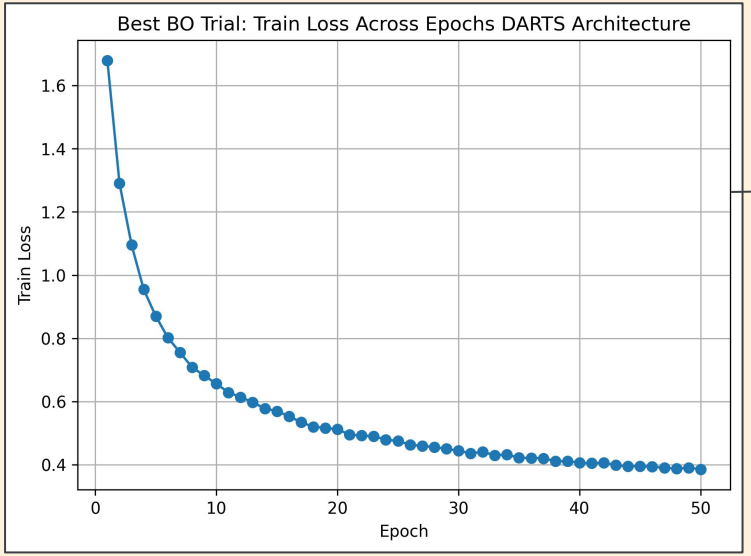


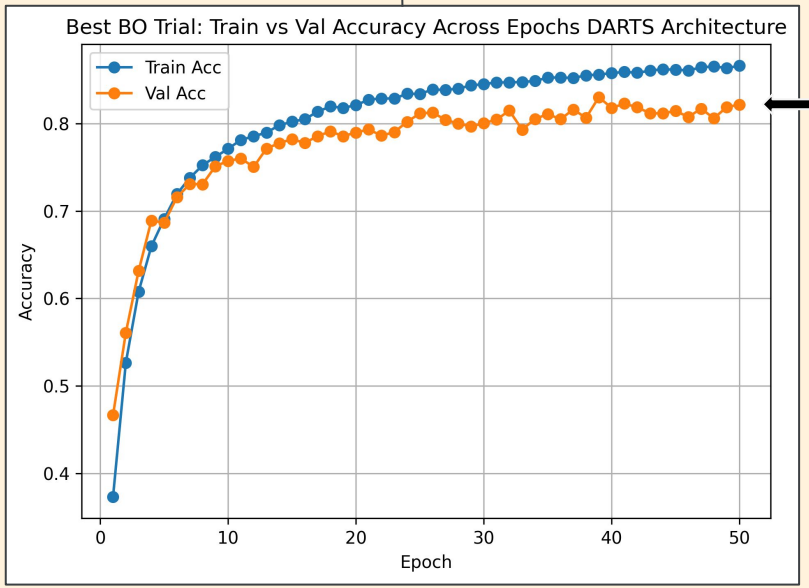
Bayesian Optimization on NAS Architecture

50 epochs, 25 trials

Bayesian Optimization on DARTS Architecture

50 epochs, 25 trials



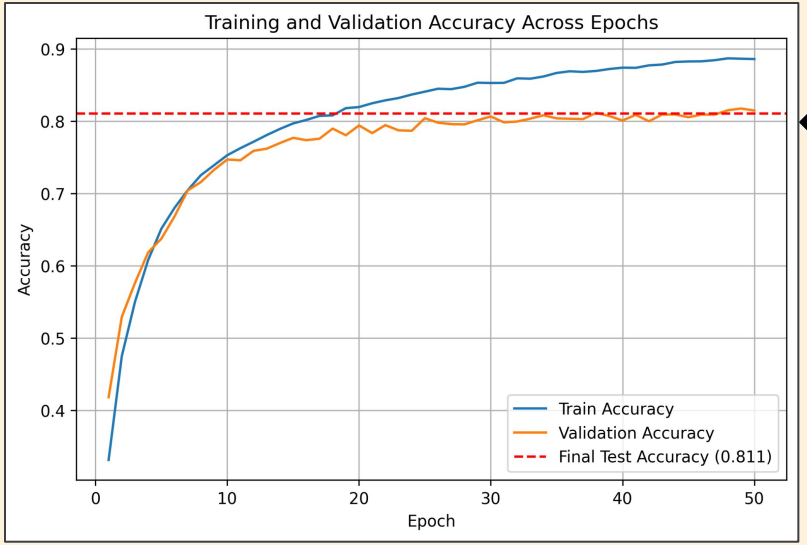


Final Val Accuracy: 83.0%

Selected Hyperparameters

Learning Rate	Weight Decay	Batch Size	Optimizer	Momentum
6.73x 10-4	4.84x 10-4	64	Adam	0.809

Results - Final Testing Data with Selected Architecture and Hyperparameters

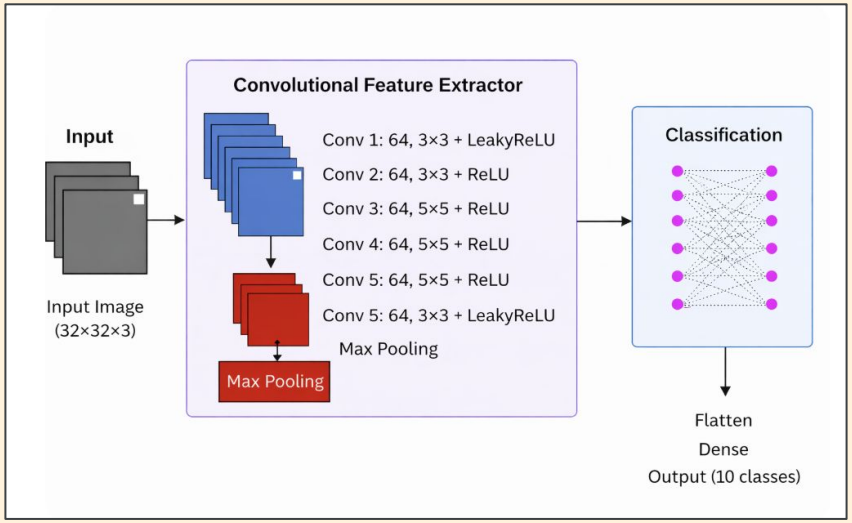


Final Test Accuracy: 81.1%

Selected Hyperparameters

Learning Rate	Weight Decay	Batch Size	Optimizer	Momentum
6.73× 10−4	4.84× 10−4	64	Adam	.809

Selected Architecture Diagram



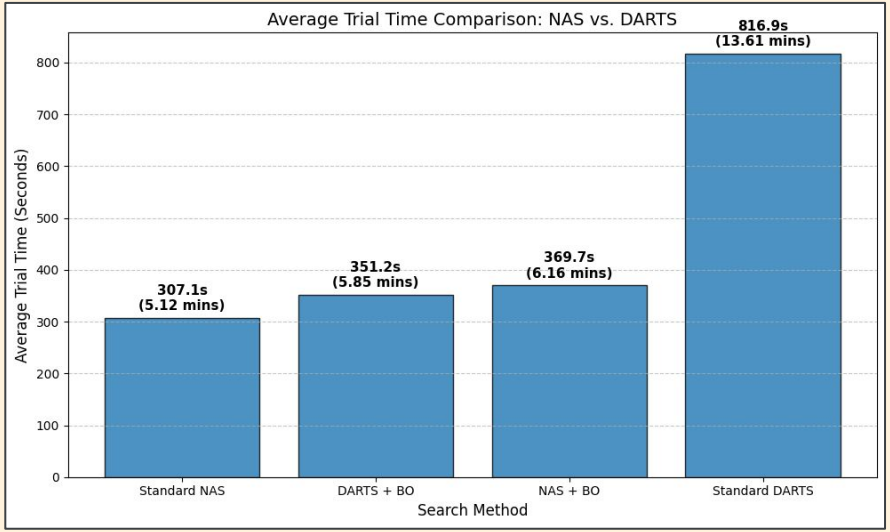
Final Model - DARTS Architecture and Corresponding Hyperparameters

50 epochs

Selected Architecture

# of Layers	Kernel Sizes	Filters	Activation Function	Pooling Type
5	[3, 3, 5, 5, 3]	All 64	[LeakyReLu, ReLu, ReLu, ReLu, LeakyReLu]	Max

Limitations



Bayesian Optimization: 25 trials, 50 epochs
Standard DARTS & Standard NAS: 50 trials, 50 epochs

Standard DARTS is much slower than NAS and DARTS with Bayesian Optimization

- Limited computational resources
  - Using random search instead of grid search which would test every possible combination
- Limited BO tuning space
- Local optimal configuration vs global optimum (50 architecture trials, 25 Bayesian optimization trials)

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